

Some Structural Relationships Bearing upon Full Employment

By Benjamin Graham

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The full-employment commitment, which was greeted skeptically by many of us in 1946, seems now on the way to becoming an accepted premise of politics and economics. Both major parties acknowledge government's responsibility for preventing continued mass unemployment.

Our earlier concern as to how this engagement can be fulfilled is no longer so manifest. The excellent employment record of the past thirteen years—and particularly the resistance shown by our economy to the depressive influences of 1954—has given American business a new confidence in its ability to keep forging ahead without nasty setbacks of the old-fashioned kind.

This optimistic thesis might be restated by some in different terms, but with the same consequences, namely: Either business will expand indefinitely at the accelerating rate required to maintain full employment, or else the Administration (of either party) will intervene as massively as is necessary to rescue us from any tailspin. This view would hold, in other words, that active business and nearly full employment are guaranteed; the only question is whether we shall have them with or without further inflation. It is probable that the bull market of 1953–55 has been reflecting the gradual permeation of investors' minds by this revolutionary conception of the future.

Before a security analyst or other economic student decides to accept or reject this crucial hypothesis, he might do well to ponder over some long-term economic relationships, which have had a significant bearing on the question of full employment in the past. We have now available reasonably dependable aggregate figures, covering the four basic factors that together determine changes in the number of unemployed: (1) productivity, (2) percapita product (or consumption), (3) hours of work, (4) labor force. These four elements are mathematically joined in the following equation:

$$\text{Unemployment} = \text{labor force} - \frac{\text{gross national product (consumption)}}{[\text{GNP per man-hour (productivity)} \times \text{average annual hours per worker}]}$$

In this article I propose to discuss some rather striking arithmetical relationships, which have existed among the four enumerated factors in the past, and then to compare the record with a recent authoritative projection to the year 1965. The source of this projection and of most of our statistical material is a recent study by the staff of the (Congressional) Joint Committee on the Economic Report, entitled "Potential Economic Growth of the United States during the Next Decade" (October 1954).

For our purpose, it has been convenient to examine the percentage changes in the four factors over four approximately equal intervals between 1900 and 1953, namely: 1913 versus 1900, 1927 versus 1913, 1940–41 average versus 1927, and 1953 versus 1940–41. None of the years taken marked a depression period, as measured by the general business index of the Cleveland

Trust Company. However, 1953 would stand out as a year of business boom, whereas the other reference years show activity moderately above or below normal, as follows:

1900	+ 3%
1913	+ 4½
1927	+3
1940–41	– 1½
1953	+15½

It follows from our previously stated formula that the maintenance of full employment depends on the interaction of three basic arithmetical relationships:

1. *The Productivity-Product Ratio.* This is the ratio of the rate of increase in productivity (GNP per man-hour) on the one hand and in consumption or living standard (GNP per capita) on the other. It produces intermediate result I, which is the change in man-hours required (worked) per unit of population.

2. *The Change in Hours of Work.* This is a composite quotient, found by dividing total man-hours worked by the number of employed. Combined with result I, it produces intermediate result II, which is the change in workers needed (employed) per unit of population.

3. *The Change in the Labor Force.* The significant figure here is the labor force in relation to population. Combined with result II, it produces final result III, which is the change in unemployment per unit of population.

Our Table 1 sets forth the basic data used in the study and some resultant figures of importance. In Table 2, the relationships previously discussed are developed as percentage changes for each of the four intervals selected. In Table 3, approximate average percentages are worked out, and these are compared in turn with the projected changes for the twelve years 1953–65 as estimated in the staff report.

Table 1

Basic Economic Data Used

	1900	1913	1927	Average 1940–41	1953	Projection 1965*
Population, millions	76.1	97.2	119.0	132.8	159.6	190.0
Adjusted population, millions**	63.8	82.7	102.1	117.5	138.1	163.5
Gross natl. product in 1953 dollars, billions	75.6	113.1	160.5	220.1	364.9	535.0
Avg. work week, hours	58.0	51.4	48.1	42.9	40.6	37.1
Working force, millions	29.1	38.7	46.9	56.9	67.0	79.0
Employment, millions	27.7	37.6	45.0	50.0	65.5	76.0
Unemployment, millions	1.4	1.1	1.9	6.9	1.5	3.0
Total man-hours worked, billions	83.5	100.4	112.6	111.5	138.2	146.6
Productivity, GNP per man-hour, 1953	0.905	1.126	1.461	1.974	2.640	3.649

Dollars

Per-Capita Adjusted

GNP, 1953 dollars	1,185	1,367	1,570	1,873	2,643	3,272
Man-hours worked	1,308	1,214	1,103	949	1,001	897
Labor force	0.456	0.468	0.460	0.484	0.485	0.483
Employment	0.434	0.454	0.442	0.426	0.474	0.465
Unemployment	0.021	0.14	0.18	0.058	0.011	0.018

* Joint Committee Staff Report, October 1954.

** Children under fourteen counted as one-half.

Table 2

Percentage Changes between Period Dates*

	1913 vs. 1900	1927 vs. 1913	1940–41 Avg. vs. 1927	1953 vs. 1940–41 Avg.
Productivity, GNP per man-hour	+19.6% h	+22.9% h	+26.0% h	+25.0% h
Product or consumption, GNP per capita adjusted	+13.4	+12.9	+16.2	+29.1
Man-hours worked**	-7.2	-9.3	-14.0	+5.2
Work week	-11.5	-6.3	-10.8	-5.4
Employment**	+4.6	-2.9	-3.6	+10.1
Labor force**	+2.8	-1.7	+4.9	+0.2
Unemployment**	-38	+28	+69	-81

* Percentage increases are calculated against higher figure. See text.

** Per capita adjusted.

Table 3

Projected Changes 1953 – 65 versus Average Changes in the Past

	Approx. Avg. Change per Past Period in 1900–53*	Projected Change, 1965 vs. 1953
Productivity	+22% h	+27.5% h
Product or consumption**	+16½	+17.9
Man-hours worked**	-6	-10.4

Work week	-8	-8.6
Employment**	+2	-1.9
Labor force**	+1½	-0.5
Unemployment**	—	+3.0

* Adjusted to twelve years.

** Per capita adjusted.

The reader should be warned that our calculations have two unusual characteristics, which we believe will make them more meaningful. Our per-capita figures are based on “adjusted population,” which counts children under fourteen as half-persons. The purpose of this device is to allow in our comparisons for the significant decline in the proportion of such children, for example, from 32.3% of total population in 1900 to 23.1% in 1940.

Second, our percentage changes are related to the “higher figure”—which means, for example, that an advance from 100 to 200 is a rise of 50 “% h,” just as a decline from 200 to 100 is a fall of 50%. This technique has interesting advantages, particularly in successive comparisons.

When we examine the percentage changes shown in the various periods we are impressed by certain interesting structural relationships, as follows:

1. Productivity has advanced throughout the fifty-three-year period at an accelerating rate. Since increased productivity by itself—i.e., if not offset by other factors—reduces the need for workers, this aspect of the employment picture has tended to present increasing difficulties.
2. Per-capita GNP, or consumption, has also increased in each period, at a fairly constant rate. Although the big upsurge in 1940–53 is most encouraging, it is exceptional in its magnitude. A good deal of it is due to the rise in the level of business, from slightly below normal in 1940–41 to a record boom activity in 1953.
3. The figures important to employment are not productivity changes or consumption changes, considered separately, but their comparative rates of growth— what I have called the productivity-product ratio. In three periods out of the four, productivity increased faster than product (or consumption), yielding a “productivity excess.” In a similar study, published in 1947, covering a greater variety of periods, I found a “productivity excess in all intervals between years of approximately similar business activity.¹

Where there is a productivity excess, the necessary result is to reduce man-hours worked per population unit. Between 1900 and 1953, this figure fell from 1,308 hours to 1,001 hours, per-capita adjusted. If the work week had not been shortened, the consequence would have been a decline of 23% in per-capita employment, producing some 16 million more unemployed in 1953.

4. The important offset to the productivity excess has, of course, been shorter hours. The work week has declined in every period, but at a somewhat irregular rate. There is no indication that hours of work have been consciously adjusted downward in an amount calculated to maintain the level of employment in good times, but the actual results have not been too far from such an objective.

The important question here is whether the productivity excess is a structural characteristic of our American economy. If there is a built-in tendency for productivity to increase faster than consumption—and the half-century figures point strongly to that conclusion—then the gradual reduction of the work week (or, better, the work year) becomes mathematically necessary to prevent a growth in unemployment.

5. Changes in labor force relative to population appear to be independent of other factors, and they have an independent effect on unemployment. Over the past fifty-three years, the number of workers has increased at a slightly greater rate than the population, that is, from 407 to 420 per 1,000 unadjusted, and from 456 to 485 per 1,000 adjusted. This difference alone, if not offset by other factors, would have increased 1953 unemployment by several millions.

There is something perverse about this expansion in our labor force. One might have expected that the tremendous advance in scale of living would have reflected itself in some decline in the working portion of our population. (Actually the influx of women into jobs has more than offset both the elimination of child labor and the trend toward earlier retirement.) The embarrassing aspect of the change in labor force is that it was concentrated almost entirely in the 1927–40 period, when it greatly aggravated the unemployment situation produced by the productivity excess.

6. Finally we must recognize that unemployment—the final and key figure—represents at bottom a marginal relationship among the four determining factors of (a) increasing productivity, (b) increasing consumption, (c) falling hours of work, and (d) variations in the labor force. The productivity increase, taken by itself, could create a serious unemployment problem in only two years; the average productivity excess would create such a problem in six years; a perverse change in the labor force could presumably do so in twelve years.

From the theoretical standpoint, at least, this is a major objection to the full-employment commitment. It could conceivably require strenuous governmental intervention in the economy when a relatively slight imbalance appears between the major factors determining employment and unemployment.

This fact is pointed up by the developments between 1952 and 1954. (The data are not given in our tables.) In these two years, real product per capita remained about unchanged, but average unemployment nearly doubled, rising from 1,673,000 to 3,200,000. The controlling elements were a rise of 4.4% in productivity, offset in part by declines of 1% each in hours of work and relative working force.

Implications for the Future

The projections into 1965, made by the staff of the Joint Committee on the Economic Report, have an interesting bearing on the previous analysis—and possibly vice-versa. Their study concluded that the economy could reach a satisfactory balance ten years hence, on the basis of a gross national product of \$535 billion—47% greater than in 1953—with no change in the price level.

But to achieve this balance, in accordance with the model presented in their study, three structural requirements will have to be met. To begin with, the staff points out, the amount of disposable income saved will have to be reduced from a recent 8 to 6%, in order to create the necessary demand for consumer goods. Second, nonmilitary expenditures by all sections of government will have to expand much more rapidly than GNP itself. Finally, there must be a

substantial further shortening of the work week; although not saying so explicitly, the discussion indicates that the alternative to such shorter hours is likely to be an inadmissible level of unemployment.

In brief, the conditions require that people work less and save less, and that civil government spend more, than at the 1953 rate. Topsy-turvy as this may sound, it is an authentic reflection of our twentieth century economy of abundance, which in turn is founded on our ever-increasing productivity.

We have worked out the percentage changes between 1953 and 1965, implied in the staff's projection, for the factors governing the level of unemployment. The figures appear in Table 3.

It will be noted that productivity is estimated to increase at a somewhat faster rate than in 1940–41 to 1953. However, the advance in consumption (per-capita GNP) is projected at a much lower rate than in 1940–41 to 1953. It is set, in fact, at about 10% more than the annual rate for the entire period 1900–53. The result is a substantial productivity excess (10.4%), and a corresponding fall in the man-hours required per capita.

The decline in the work week is anticipated (or stipulated) to be almost exactly at the average rate for the half-century. The composite result of the three factors is a decline of just under 2% in per-capita employment, one-quarter of which is offset by a minor shrinkage in relative labor force. The final calculation shows an increase in unemployment from 11 to 18 per 1,000 population, adjusted.

This is, of course, a very encouraging projection. (But the staff warns us that it presents rather a picture of how maximum economic growth may be achieved than predictions of actual developments.) It is certainly not Utopian, for it appears to be quite closely in line with the secular growth of our economy, as analyzed in our four-period comparisons.

But, even if the estimates impress us as reasonable, two caveats should be borne in mind by those studying the figures as a possible guide to investment and business decisions. The first is that these satisfactory twelve-year advances are not in themselves incompatible with a substantial interim setback. In each of the three periods during 1900–41 we had business depressions, although the economy showed rather consistent growth between all the terminal years.

The prevention of a cyclical downturn, with resultant high unemployment, is a separate problem. Will the economy be able to resist depressive forces in the future as well as it did in 1953–54? I doubt whether our analysis, or that of the Joint Committee's staff, sheds any definitive light on this important question.

The second caveat brings us back to the marginal nature of the unemployment total. It would take a comparatively small adverse change in the first three of our controlling factors to change the final figure of unemployment from a fairly satisfactory to a problem-making figure. For example, if the work week should fall 2 hours instead of 3½, the arithmetical result would be to double the unemployment figure—i.e., to 6 million. Nor is it impossible that the fourth factor—labor force—may add substantially to difficulties created by the other elements.

The American economy has been doing remarkably well in recent years. But it would be premature to conclude that it is now both depression-proof and unemployment-proof. Nor are the two characteristics necessarily the same.

All data for 1913 and later years are taken from the Joint Committee's Report. Man-hours employed have been raised to allow for general government employees, excluded in their figures. The 1900 data came from various sources, selected to accord most reasonably with the Joint Committee's series.

Use of the "Percent of High" Convention

Percentage increases are calculated in this article against the higher or terminal figure. I have long believed that this convention is better suited to the work of economists and security analysts than the standard method of calculating percentage gains against the starting figure. I am emboldened to employ it here because of its use in a study published by the Department of Economic Affairs of the United Nations entitled "Instability in the Export Markets of Underdeveloped Countries" (1952). As pointed out in Appendix B of that paper, the "percent of high" or "% h" convention (my designations) has several advantages over the standard method, namely:

1. By the "% h" method the series 100, 80, 100 shows a fall of 20%, followed by a rise of 20%, instead of the conventional (and troublesome) 20% fall and 25% rise.
2. A single rise is always under 100% h, corresponding to the maximum single decline (where positive numbers only are used).
3. In two series such as 10, 5, 8, 10 and 5, 10, 8, 5 the average absolute fluctuations are the same under the "% h" method, whereas they differ markedly under the standard method.

Note that "percent of high" is recommended only for comparing changes between successive values. It is not applicable to calculating percentage of profit mortality and the like.

Notes

1. See Benjamin Graham, "National Productivity: Its Relationship to Unemployment in Prosperity," *American Economic Review*, vol. 37 (May 1947), pp. 384–396.